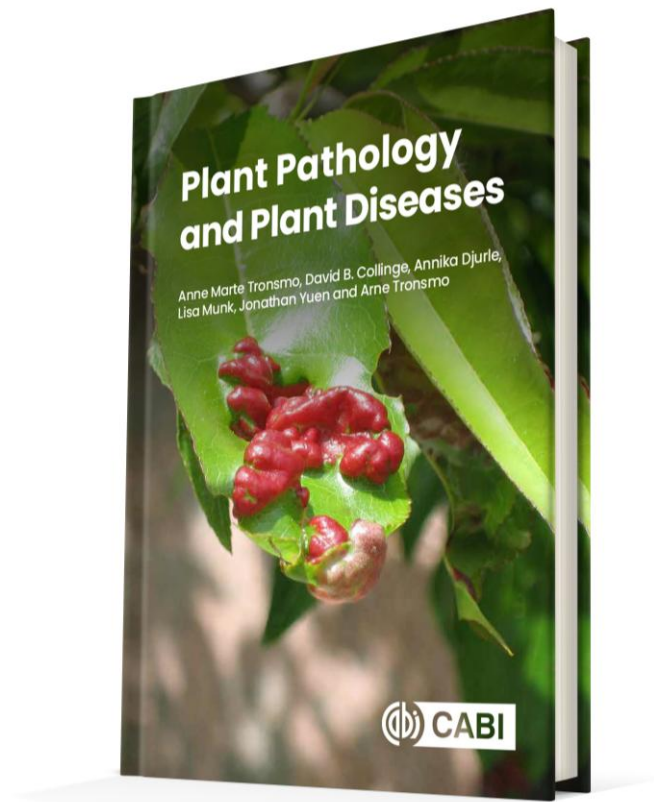




Plant Pathology and Plant Diseases

© Anne Marte Tronsmo, David B. Collinge, Annika Djurle,
Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

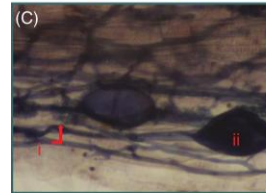
Fig. 3.1 A-E
Examples of lifestyles
exhibited by fungi
associated with plants: (A)
 Endophytic; (B) Pathogenic
 biotrophic; (C) Symbiotic
 biotroph; (D)
 Hemibiotrophic; (E)
 Necrotrophic. (A,B,D,E ©
 D.B.Collinge C: © Eva Lucic.)



Endophyticism/endophytic
 The microorganism lives inside the plant without causing disease. Its presence may even benefit the plant by increasing abiotic stress tolerance or suppressing other microorganisms.
 An endophyte growing from a wheat kernel.



Pathogenic biotroph/biotrophic
 Obtains nutrition from the living plant and harmful as it acts as a sink for nutrients which the plant needs to grow.
Phragmidium spp. – one of several rusts on rose. Other examples: powdery mildews, downy mildews, viruses, *Agrobacterium*. Many of these are obligate parasites.



Symbiotic biotroph/biotrophic
 The interaction is obligate. The microorganism provides nutrients (e.g., phosphorus) in return for nutrition.
 Vesicular Arbuscular mycorrhiza (*Rhizophagus irregularis*)
 i: arbuscule: a highly branched hyphal structure for nutrient exchange. ii: spore.



Hemibiotrophy/hemibiotrophic
 Traditionally the first phase is considered to be biotrophic, followed by necrotrophy. Many are probably endophytic followed by a necrotrophic phase. *Zymoseptoria tritici*, an ascomycete on wheat: This pathogen has an asymptomatic (endophytic) period of a couple of weeks before switching to necrotrophy, where macroscopic symptoms can be observed. *Phytophthora* spp. (Oomycete) can have a true biotrophic phase and switch to necrotrophy.



Necrotrophy/necrotrophic
 Kills the host and consumes the nutrients in the dead tissue. *Rhizoctonia solani sensu lato* is a necrotrophic basidiomycete with a broad host spectrum, here on rice.
 Other common examples: *Botrytis* spp., *Sclerotinia* spp.
 Bacteria: *Pectobacterium* spp.

Fig. 3.1 F – I
Examples of lifestyles
exhibited by fungi
associated with plants: F)
Saprophytic; (G) Facultative
symbiosis; (H) Parasitic; (I)
Epiphytic. (© D.B.Collinge.)



Saprophytism/saprophytic
Consumes nutrients in dead plant tissue. Also called
saprotrophic. The black structures are the signs of
saprophytic fungi such as *Cladosporium herbarum* or
Alternaria spp.



Facultative symbiosis /symbiotic
Can switch between saprophytism and symbiosis.
Amanita muscaria is a common ectomycorrhizal
basidiomycete typically found in association with birch
(*Betula* spp.).
Bacteria: Rhizobia.



Parasiticism/parasitic
One organism is entirely dependent on the other. Many
biotrophic pathogens are parasites. *Monotropa*
hypopitys – Dutchman's pipe or Bird's nest is a plant
which parasitizes ectomycorrhizae which are in turn
facultative symbionts with their hosts.



Epiphyticism/epiphytic
Uses the plant as a physical substrate on which to grow.
Growth on the leaf surface - the
phyllosphere, or associated with the root -
the rhizosphere.
An epiphytic lichen on a leaf. Lichens are themselves
symbionts between algae and fungi. The combined
organism is an epiphyte in this example.

Fig. 3.2 Schematic illustration of a generalized disease cycle. (©A.M. Tronsmo.)

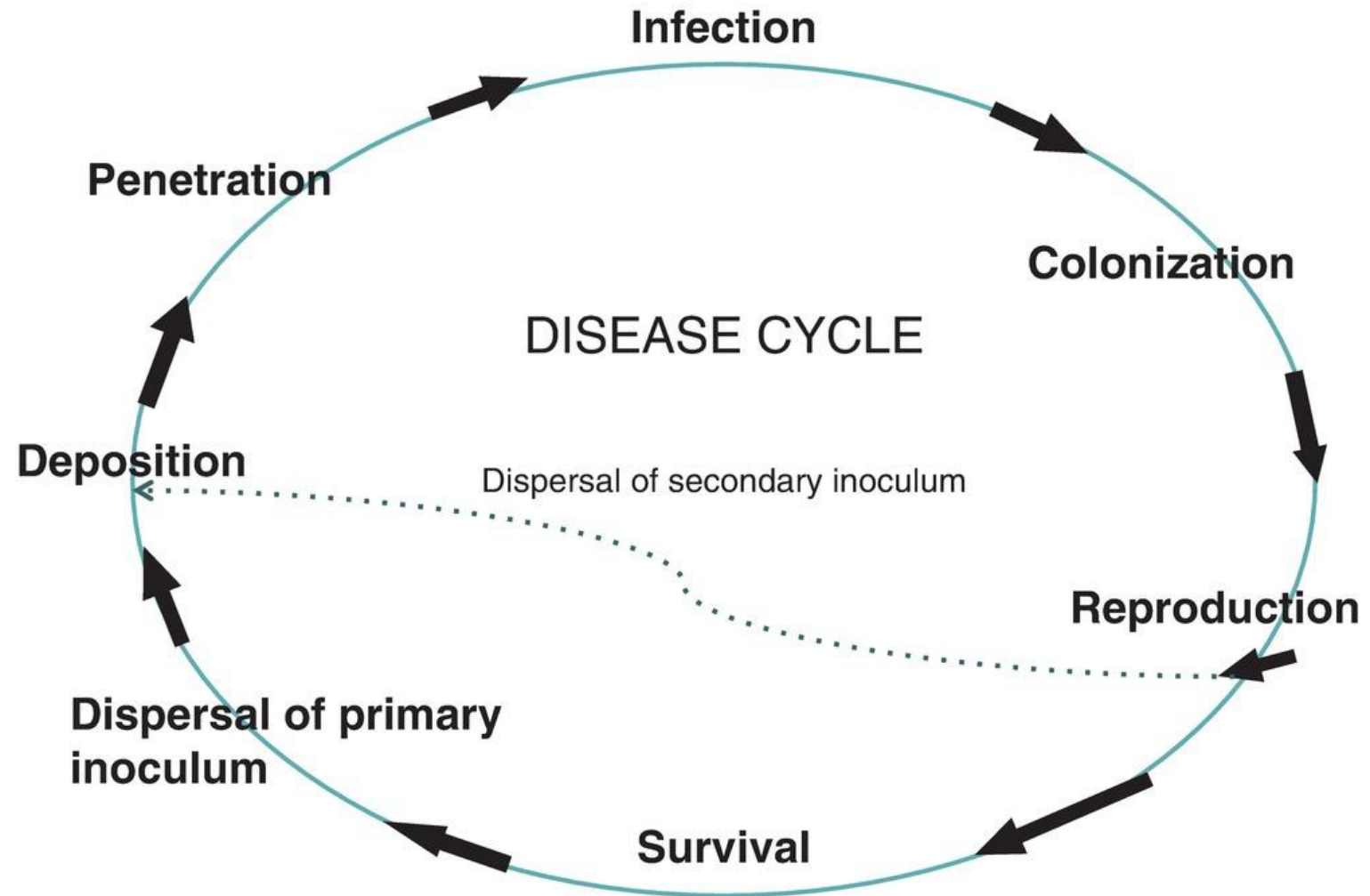


Fig. 3.3 Stages in a disease cycle. A and B: Arrival of inoculum; C: Germinating spore; D: Formation of appressorium, infection peg (a specialized hypha) and cuticle penetration; E: The pathogen procure nutrients from the host via a haustorium (feeding hypha); F: Colonization of the host tissue. (© A.M. Tronsmo.)

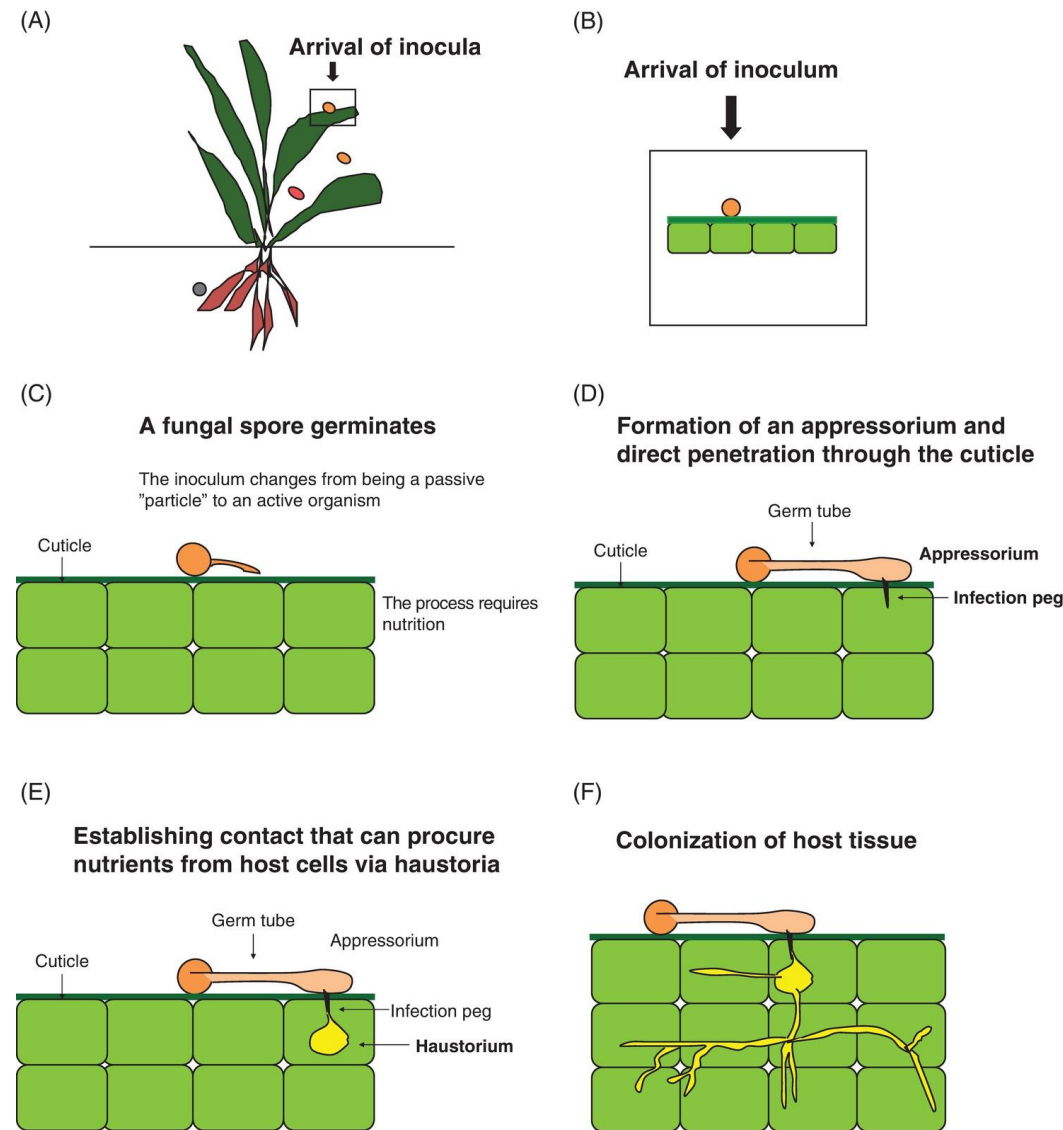


Fig. 3.4
Illustrations of haustoria
from fungal and oomycete
pathogens. (A)
Hyaloperonospora parasitica
 causing downy mildew in
 broccoli; (B) *Blumeria*
graminis causing powdery
 mildew in a barley epidermal
 cell; (C) *Hemileia vastatrix*
 causing coffee rust and
Puccinia striiformis causing
 yellow rust on wheat. (A: ©
 B. Dahl Jensen, B: © Y. Wei &
 C: C. K. Sørensen; Drawings
 © D.B.Collinge.)

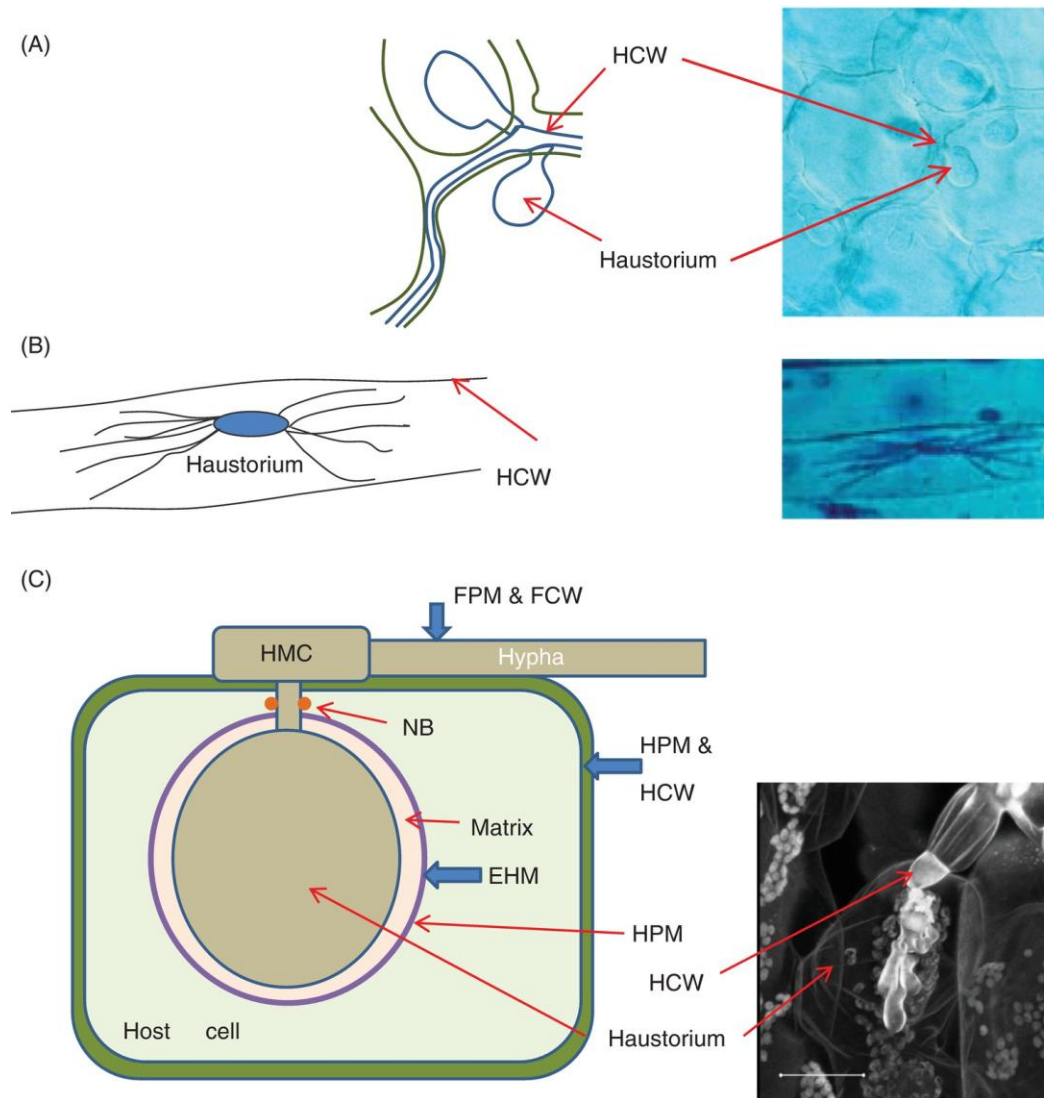
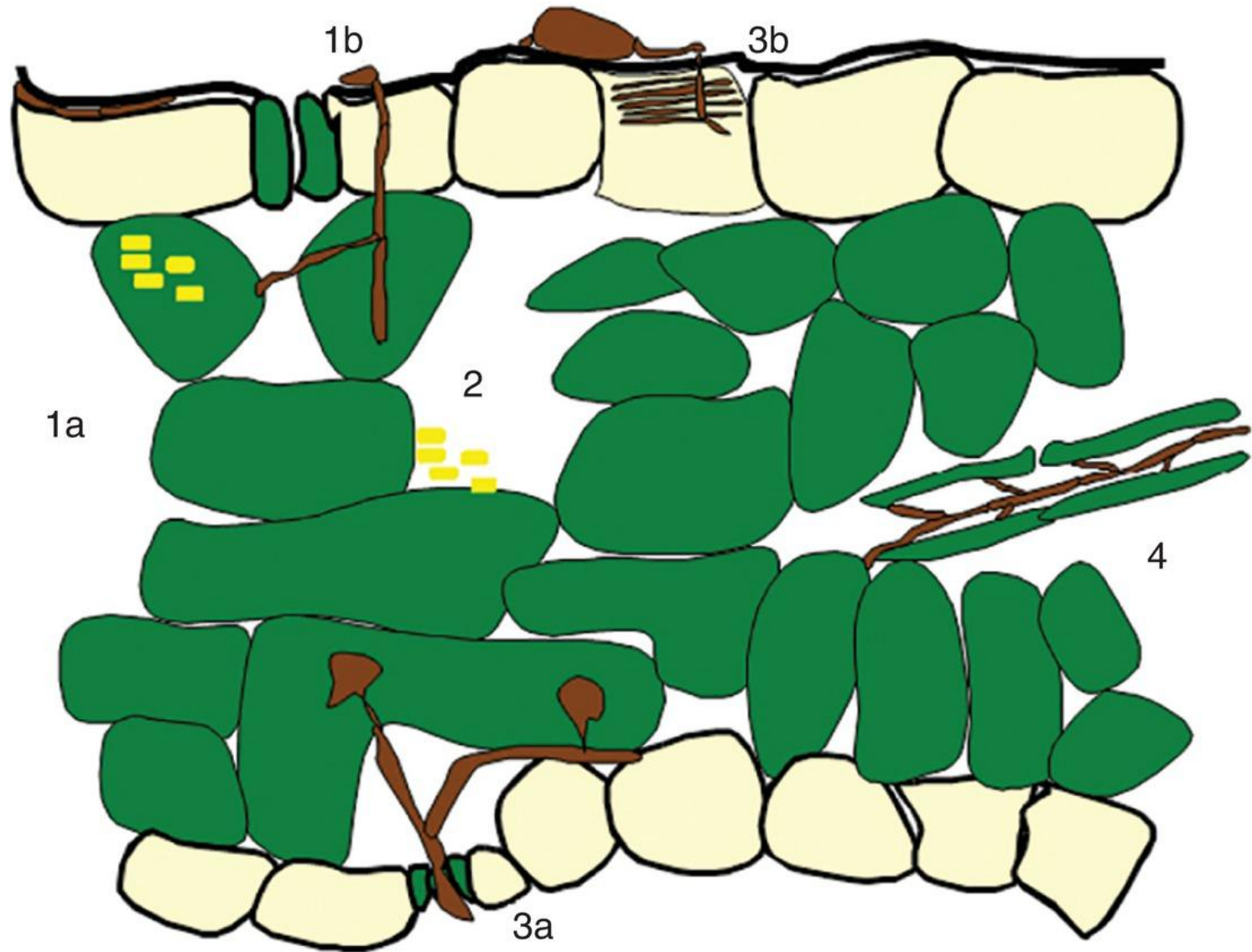
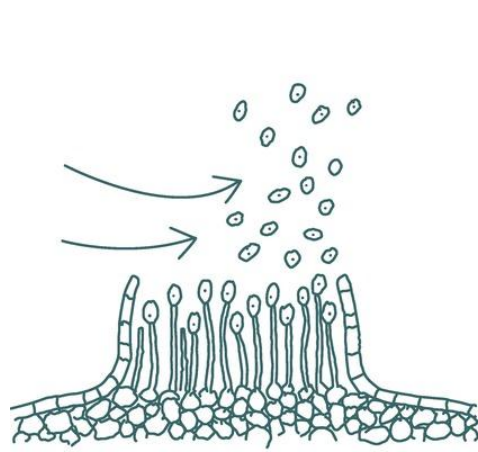


Fig. 3.5 Choice of niche. (1a) Intracellular (e.g. virus). (1b) Invasively – direct penetration; (2) Intercellular (e.g. bacteria); (3) Intercellular with haustoria entry via stoma (3a, e.g. rust) or essentially epiphytic with haustoria (3b, e.g. powdery mildew); (4) Vascular (e.g. vascular wilt bacteria and fungi, phytoplasma). (© D.B. Collinge.)



**Fig. 3.6 Dispersal of
plant pathogens.**

Drawing H. Karlsen
(© NIBIO.)



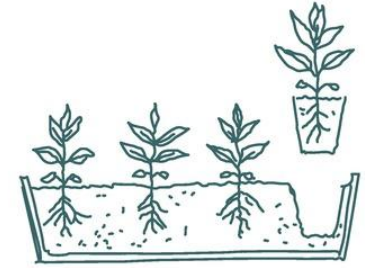
Wind



Rain



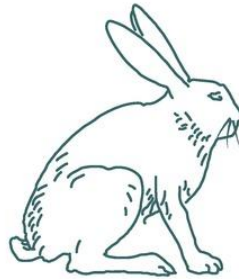
Infected seed



Infected plants



Insects



Animals



Humans, tools and machinery

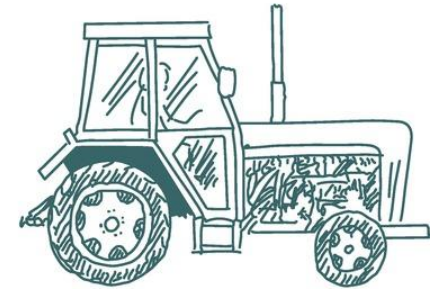


Fig. 3.7

A Some means of survival of plant pathogens. (© NIBIO.)

B Structures for survival of fungi and oomycetes.

(Drawing H. Karlsen © NIBIO.)

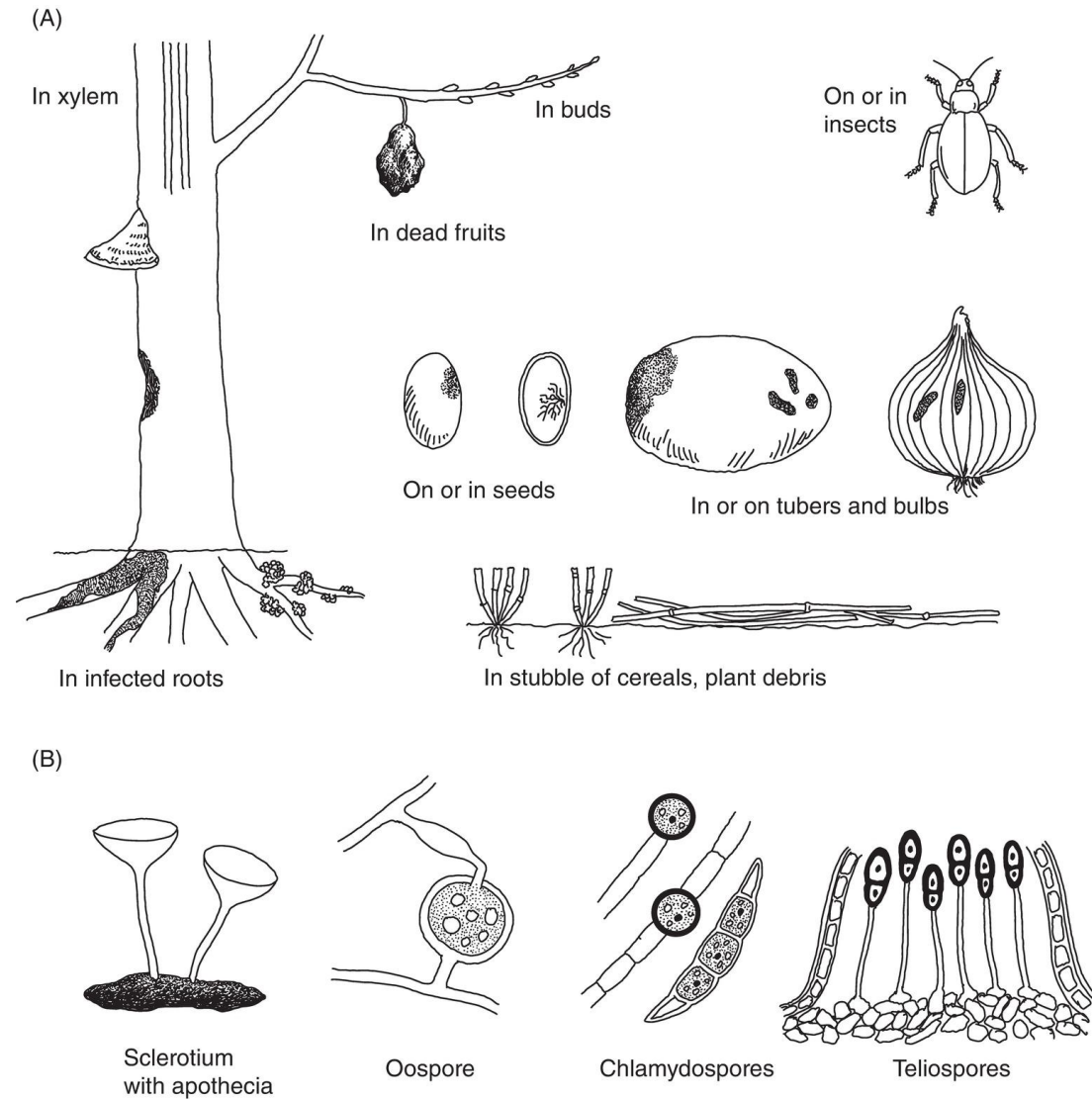


Fig. 3.8 Example of a disease cycle for a monocyclic pathogen, *Sclerotinia sclerotiorum*. (Drawing H. Karlsen, © NIBIO.)

